

Delivery Route Optimizer

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The **Delivery Route Optimizer** is an intelligent logistics solution designed to improve the efficiency of daily delivery operations by generating near-optimal routes for multiple vehicles. The system begins by cleaning and standardizing inconsistent address and stop data to ensure accurate route planning. It then assigns delivery stops to vehicles based on factors such as vehicle capacity, delivery locations, and customer time windows. Advanced optimization algorithms are used to minimize total travel distance, fuel consumption, and delivery time while maximizing vehicle utilization. Since exact solutions become computationally expensive for large-scale delivery networks, the system employs approximation and heuristic techniques to produce high-quality routes within practical time limits. Additionally, the engine provides comprehensive cost and quality reports, including total distance travelled, estimated travel time, fuel usage, route efficiency, and constraint satisfaction. This project helps logistics companies reduce operational costs, improve on-time deliveries, enhance customer satisfaction, and support scalable, data-driven transportation planning in real-world environments.

Signature of the Guide